

# Alignment of Genome Sequencing Data

## GSND 5340Q, BMDA

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# Sequencing data alignment

Next-Generation Sequencing (NGS) Data present new challenges:

- ▶ Map 'reads' to genome
- ▶ Call SNPs and variants from the reads
- ▶ Other Applications: RNA-seq, miRNAs, alternative splicing, ChIP-seq, BS-seq, RNA editing

# Sequencing data alignment

Obstacles:

- ▶ Massive data size
- ▶ Repeat regions, rare variants

**Goal:** Develop an approach to put the puzzle together!

# Mapping Algorithms

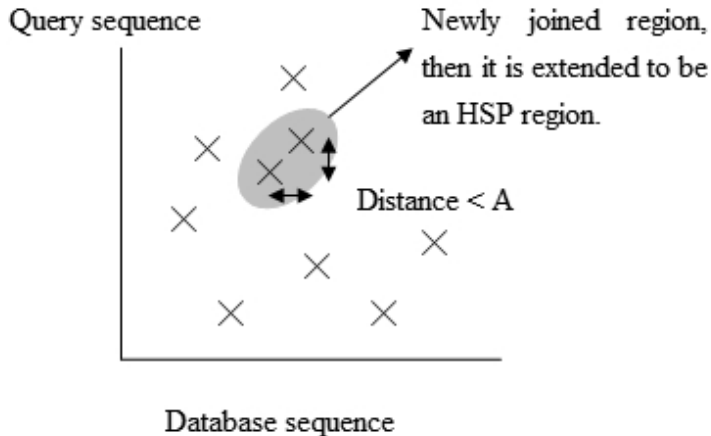
Alignment algorithms (not comprehensive – some of my favorites!)

- ▶ BLAST
- ▶ BLAT
- ▶ SOAP2
- ▶ Bowtie2
- ▶ BWA
- ▶ STAR
- ▶ Subread (Rsubread)
- ▶ GNUMAP
- ▶ NovoAlign
- ▶ Many others!!

# Basic Local Alignment Search Tool (BLAST)

BLAST is a “heuristic” method that identifies and combined *High-scoring Segment Pairs (HSPs)* between two sequences. BLAST searches for high scoring sequence alignments between the query sequence and the existing sequences in the database using an approach that approximates the Smith-Waterman algorithm (described later).

# Basic Local Alignment Search Tool (BLAST)



# Basic Local Alignment Search Tool (BLAST)

BLAST can be found at:

<https://blast.ncbi.nlm.nih.gov/Blast.cgi>BLAST. Try “BLASTING” the following sequences:

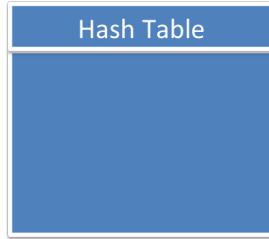
```
>NC_045512.2 Severe acute respiratory syndrome coronavirus 2 isolate Wuhan-Hu-1  
ATTAAAGGTTTATACCTTCCCAGGTAACAAACCAACCAACTTTTCGATCTCTTGTAGATCTGTTCTCTAAACGAACTTTAA
```

```
>NC_001802.1 Human immunodeficiency virus 1  
GGTCTCTCTGGTTAGACCAGATCTGAGCCTGGGAGCTCTCTGGCTAACTAGGGAACCCACTGCTTAAGCCTCAATAAAGC
```

```
>NR_102810.2 Mycobacterium tuberculosis H37Rv 16S ribosomal RNA  
TGTTTGGAGAGTTTGATCCTGGCTCAGGACGAACGCTGGCGGCGTGCTTAACACATGCAAGTCGAACGGA  
AAGGTCTCTTCGGAGATACTCGAGTGGCGAACGGGTGAGTAACACGTGGGTGATCTGCCCTGCACTTCGG
```

```
>NR_041248.1 Bacillus anthracis strain ATCC 14578 16S ribosomal RNA  
AGAGCTTGCTCTTATGAAGTTAGCGGCGGACGGGTGAGTAACACGTGGGTAACTGCCATAAGACTGGG
```

# Creating a Hash Index Table

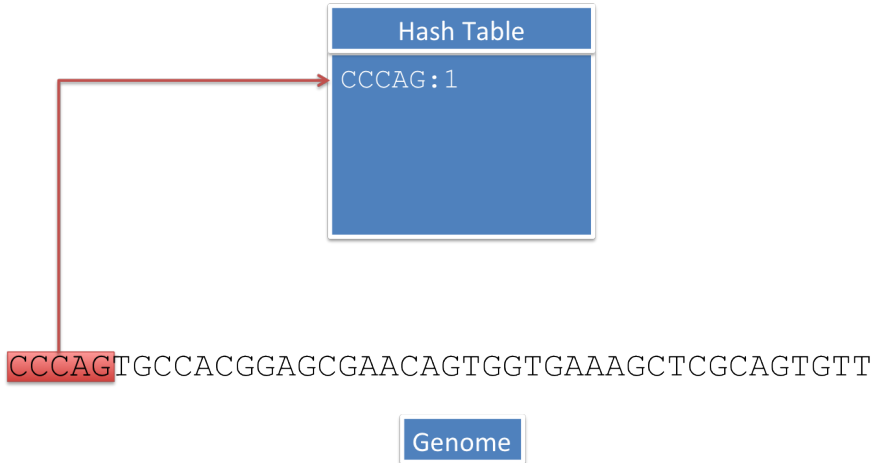


CCCAGTGCCACGGAGCGAACAGTGGTGAAAGCTCGCAGTGTT

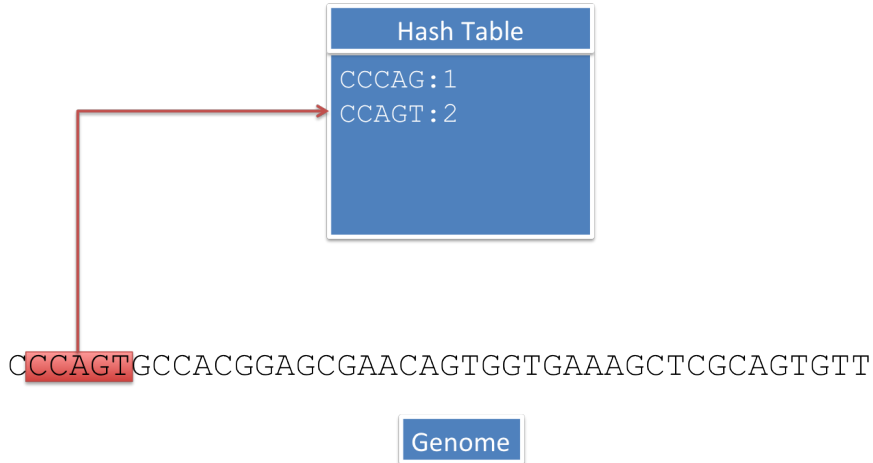
Genome



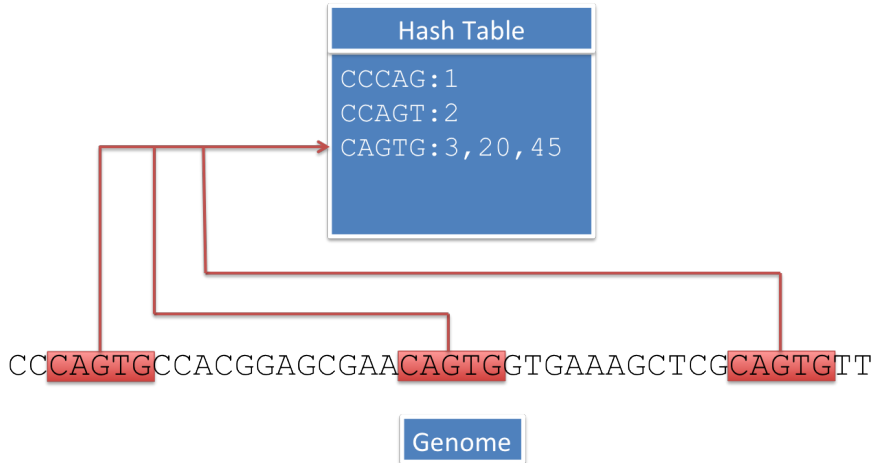
# Creating a Hash Index Table



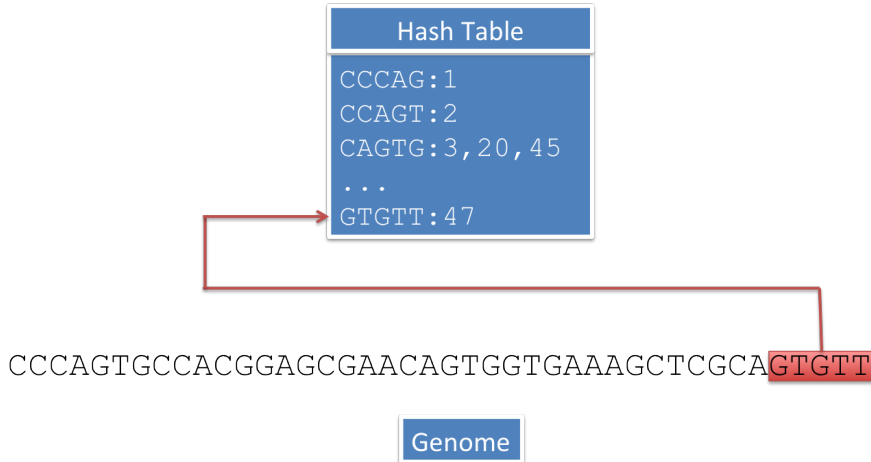
# Creating a Hash Index Table



# Creating a Hash Index Table



# Creating a Hash Index Table



# Read Lookup

Read

CAGTGCC

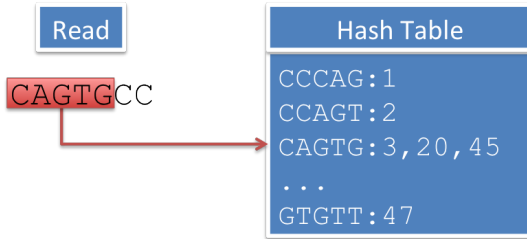
Hash Table

CCCAG:1  
CCAGT:2  
CAGTG:3, 20, 45  
...  
GTGTT:47

CCCAGTGCCACGGAGCGAACAGTGCTGAAAGCTCGCAGTGTT

Genome

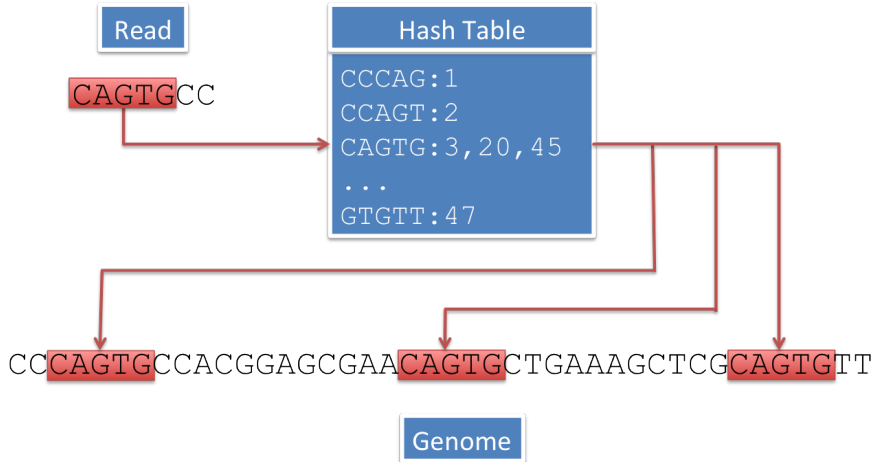
# Read Lookup



CCCAGTGCCACGGAGCGAACAGTGCTGAAAGCTCGCAGTGTT

Genome

# Read Lookup



# Read Lookup

Read

CAGTGCC

Hash Table

CCCAG:1  
CCAGT:2  
CAGTG:3, 20, 45  
...  
GTGTT:47

CAGTGCC                      CAGTGCC                      CAGTGCC  
| | | | |                      | | | | |                      | | | | |  
CC**CAGTGCC**ACGGAGCGAA**CAGTGCT**GAAAGCTCG**CAGTGTT**

Genome



# Read Lookup

Read

CAGTGCC

Hash Table

CCCAG:1  
CCAGT:2  
CAGTG:3, 20, 45  
...  
GTGTT:47

CAGTGCC

|||||

CC**CAGTGCC**ACGGAGCGAA**CAGTGC**TGAAAGCTCG**CAGTGTT**  
Score=7

CAGTGCC

|||||

Score=5

CAGTGCC

|||||

Score=3

Genome

# Needleman-Wunsch

T A C A T C

T T C A - C

|   | t | a | c | a | t | c |
|---|---|---|---|---|---|---|
| t |   |   |   |   |   |   |
| t |   |   |   |   |   |   |
| c |   |   |   |   |   |   |
| a |   |   |   |   |   |   |
| c |   |   |   |   |   |   |

- Allows for mismatches and gaps
- Scoring function: +1 match; -1 for mismatch; -2 for gap
- Produces maximum likelihood alignment

# Needleman-Wunsch

T A C A T C

T T C A - C

|   |     | t  | a  | c  | a  | t   | c   |
|---|-----|----|----|----|----|-----|-----|
| t | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| t | -2  |    |    |    |    |     |     |
| c | -4  |    |    |    |    |     |     |
| a | -6  |    |    |    |    |     |     |
| a | -8  |    |    |    |    |     |     |
| c | -10 |    |    |    |    |     |     |

- Allows for mismatches and gaps
- Scoring function: +1 match; -1 for mismatch; -2 for gap
- Produces maximum likelihood alignment

# Needleman-Wunsch

```

T A C A T C
|
T T C A - C

```

|   |     | t  | a  | c  | a  | t   | c   |
|---|-----|----|----|----|----|-----|-----|
| t | 0   | -2 | -4 | -6 | -8 | -10 | -12 |
| t | -2  | 1  |    |    |    |     |     |
| t | -4  |    |    |    |    |     |     |
| c | -6  |    |    |    |    |     |     |
| a | -8  |    |    |    |    |     |     |
| c | -10 |    |    |    |    |     |     |

- Allows for mismatches and gaps
- Scoring function: +1 match; -1 for mismatch; -2 for gap
- Produces maximum likelihood alignment

# Needleman-Wunsch

```

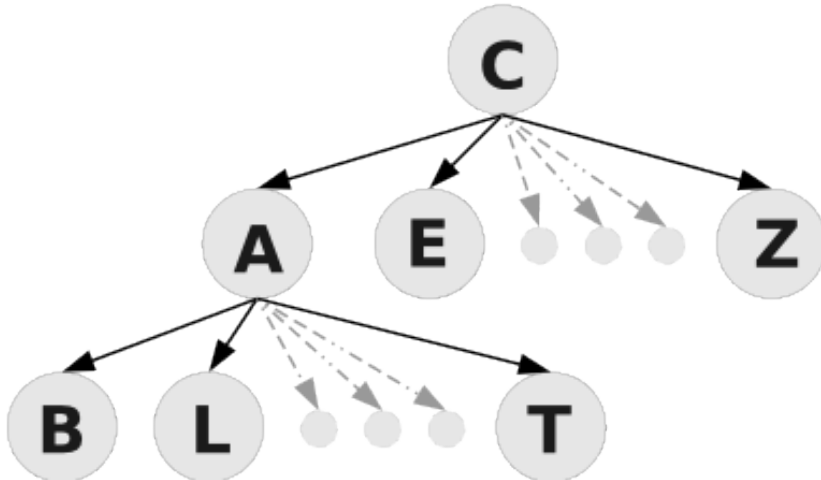
T A C A T C
|   | |   |
T T C A - C

```

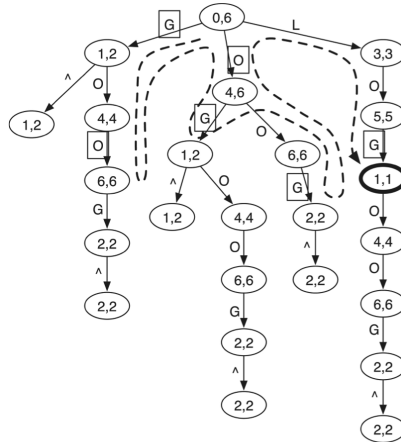
|   |  | t   | a        | c        | a        | t        | c        |          |
|---|--|-----|----------|----------|----------|----------|----------|----------|
|   |  | 0   | -2       | -4       | -6       | -8       | -10      | -12      |
| t |  | -2  | <b>1</b> | -1       | -3       | -5       | -7       | -9       |
| t |  | -4  | -1       | <b>0</b> | -2       | -4       | -4       | -6       |
| c |  | -6  | -3       | -2       | <b>1</b> | -1       | -3       | -3       |
| a |  | -8  | -5       | -2       | -1       | <b>2</b> | <b>0</b> | -2       |
| c |  | -10 | -7       | -4       | -1       | 0        | 1        | <b>1</b> |

- Allows for mismatches and gaps
- Scoring function: +1 match; -1 for mismatch; -2 for gap
- Produces maximum likelihood alignment

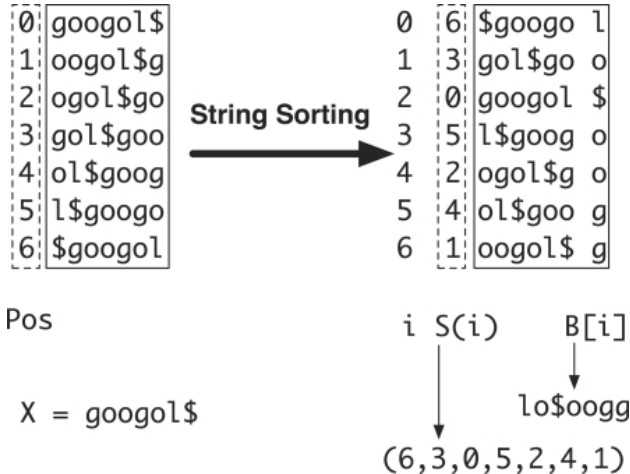
# Prefix Trees



# Prefix Trees

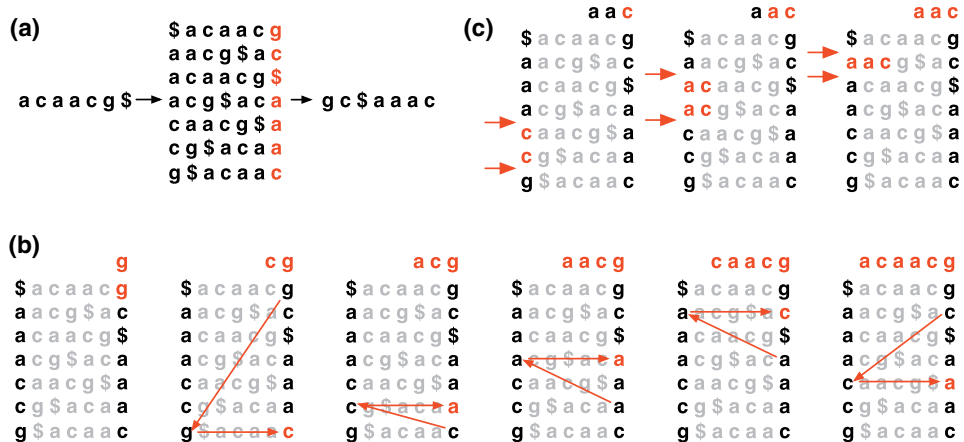


# Burrows-Wheeler Transformation (BWA, Bowtie)

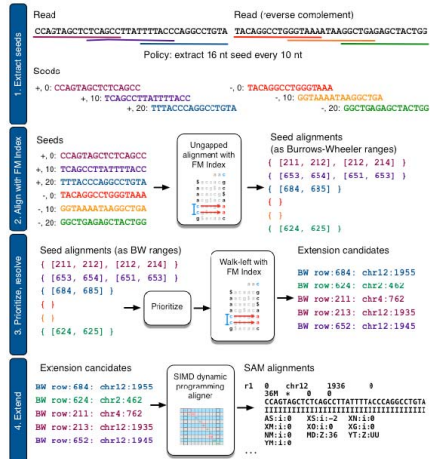




# Burrows-Wheeler Transformation (BWA, Bowtie)



# Hybrid Methods (BWA-SW, Bowtie2)



# Read alignments are stored in SAM/BAM/CRAM

- ▶ BAM and CRAM are compressed SAM files
- ▶ One line per alignment
- ▶ Use `samtools` to view, sort, merge, concatenate, index, get statistics, etc, on alignment files.

# SAM/BAM/CRAM Format – Header

The *header* lines start with @:

- ▶ @HD = header definition
- ▶ @SQ = a sequence in the reference file you used, followed by how long it was and its comment (from the reference file)
- ▶ @RG = read groups you assigned while mapping the reads
- ▶ @PG = programs used to obtain this bam, in order

# SAM/BAM/CRAM Format – Body

Each alignment has 11 mandatory fields:

| Col | Field | Type   | Regexp/Range                             | Brief description                     |
|-----|-------|--------|------------------------------------------|---------------------------------------|
| 1   | QNAME | String | [!-?A-~]{1,255}                          | Query template NAME                   |
| 2   | FLAG  | Int    | [0,2 <sup>16</sup> -1]                   | bitwise FLAG                          |
| 3   | RNAME | String | \*  [!-( )+-<>-~] [!-~]*                 | Reference sequence NAME               |
| 4   | POS   | Int    | [0,2 <sup>29</sup> -1]                   | 1-based leftmost mapping POSition     |
| 5   | MAPQ  | Int    | [0,2 <sup>8</sup> -1]                    | MAPping Quality                       |
| 6   | CIGAR | String | \*  ([0-9]+[MIDNSHPX=])+                 | CIGAR string                          |
| 7   | RNEXT | String | \* =  [!-( )+-<>-~] [!-~]*               | Ref. name of the mate/next segment    |
| 8   | PNEXT | Int    | [0,2 <sup>29</sup> -1]                   | Position of the mate/next segment     |
| 9   | TLEN  | Int    | [-2 <sup>29</sup> +1,2 <sup>29</sup> -1] | observed Template LENgth              |
| 10  | SEQ   | String | \*  [A-Za-z=.]+                          | segment SEQUENCE                      |
| 11  | QUAL  | String | [!-~]+                                   | ASCII of Phred-scaled base QUALity+33 |

# SAM/BAM/CRAM Format – Flag

The flag is the summation of the following binary attributes:

| Binary (Decimal)   | Hex   | Description                                                                                                |
|--------------------|-------|------------------------------------------------------------------------------------------------------------|
| 00000000001 (1)    | 0x1   | Is the read paired?                                                                                        |
| 00000000010 (2)    | 0x2   | Are both reads in a pair mapped “properly” (i.e., in the correct orientation with respect to one another)? |
| 00000000100 (4)    | 0x4   | Is the read itself unmapped?                                                                               |
| 00000001000 (8)    | 0x8   | Is the mate read unmapped?                                                                                 |
| 00000010000 (16)   | 0x10  | Has the read been mapped to the reverse strand?                                                            |
| 00000100000 (32)   | 0x20  | Has the mate read been mapped to the reverse strand?                                                       |
| 00001000000 (64)   | 0x40  | Is the read the first read in a pair?                                                                      |
| 00010000000 (128)  | 0x80  | Is the read the second read in a pair?                                                                     |
| 00100000000 (256)  | 0x100 | Is the alignment not primary? (A read with split matches may have multiple primary alignment records.)     |
| 01000000000 (512)  | 0x200 | Does the read fail platform/vendor quality checks?                                                         |
| 10000000000 (1024) | 0x400 | Is the read a PCR or optical duplicate?                                                                    |

Example flags:

Single end read mapped to + strand: 0 (no flags apply)

Single end read mapped to - strand: 16

Unmapped, paired end first mate read with unmapped mate:  $69 = 1 + 4 + 8 + 64$

# SAM/BAM/CRAM Format – Example

```
@HD      VN:1.0  SO:coordinate
@SQ      SN:chr20      LN:64444167
@PG      ID:TopHat      VN:2.0.14      CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
HWI-ST1145:74:C101DACXX:7:1102:4284:73714      16      chr20      190930      3      100M      *      0      0
      CCGTGTTTAAAGGTGGATGCGGTACCTTCCCAGCTAGGCTTAGGGATTCTAGTTGGCCTAGGAAATCCAGCTAGTCCTGTCTCTCAGTCCCCCTCT
C      BBDDCCDDCCDDDDDDDDDDDDDCDDCCBC?DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD
AS:i:-15      XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714  HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961      16      chr20      193953      50      100M      *      0      0
      TGCTGGATCATCTGGTTAGTGGCTTCTGACTCAGAGGACCTTCGTCCCCTGGGGCAGTGGACCTTCCAGTGATTCCCCTGACATAAGGGGCATGGACGA
G      DCDDEDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD
AS:i:-16      XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030      16      chr20      270877      50      100M      *      0      0
      GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCACCTGGCCAGCAGCACCAACCAGAAAGAAGGAAGAAGACAGGAAAAAACCA
C      DDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDDD
AS:i:-11      XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13      NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699      0      chr20      271218      50      50M4700N50M      *      0
      0      GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGTGCACTTGGGTCTCCGAAGCAGAACATCTCAAATATGACCTCTCG
```

# SAM/BAM/CRAM Format

- ▶ Use `samtools view` to see the content of a SAM/BAM/CRAM file.
- ▶ Always sort and index them
- ▶ Use `samtools view -f` (include) and `-F` (exclude) to filter by flags
- ▶ Use `samtools view -q` to filter by quality.
- ▶ More on `samtools` later!!



# BWA Example (OnDemand Amarel Desktop)

```
# load bwa
```

```
module load bwa
```

```
# Index the genome:
```

```
bwa index genomefile.fa
```

```
# Align the reads:
```

```
bwa mem genomefile.fa myfastqfile.fq > bwaalign.sam
```

```
# Note: for use with gatk (will learn later) bwa mem
```

```
# should include a `read group' header as well, e.g.:
```

```
-R "@RG\tID:group1\tSM:sam1\tPL:illumina\tLB:lib1\tPU:unit1"
```

# Bowtie2 Example (OnDemand Amarel Desktop)

*# Load Bowtie2*

```
module load bowtie2
```

*# Index the genome:*

```
bowtie2-build genomefile.fa output/genomefile
```

*# Align the reads:*

```
bowtie2 -x output/genomefile -U myfastqfile.fq \  
-S bowtiealign.sam
```

# RSubread Example (OnDemand RStudio)

```
# Install the Bioconductor installer
install.packages("BiocManager")

# Install Rsubread
BiocManager::install("Rsubread")

# Index the genome:
buildindex("genomefile", "genomefile.fa")

# Align the reads:
align("genomefile", "myfastqfile.fq",
      output_file="myfastq.bam", nthreads = 4)
```

# Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Denver
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.5.2    fastmap_1.2.0     cli_3.6.5         tools_4.5.2
## [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.17.1 yaml_2.3.12
## [9] rmarkdown_2.30    knitr_1.51        xfun_0.55         digest_0.6.39
## [13] rlang_1.1.6       evaluate_1.0.5
```